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Author(s):
Anderegg, Benjamin; Uhlmann, Florian; Sornette, Didier

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The impact of option hedging on the spot market volatility

Benjamin Anderegg [a], Florian Ulmann [b,*], Didier Sornette [b,c,d]

[a] Swiss National Bank, Boersenstrasse 15, 8001 Zuerich, Switzerland
[b] ETH Zurich, Department of Management, Technology, and Economics, Scheuchzerstrasse 7, 8092 Zurich, Switzerland
[c] Swiss Finance Institute, c/o University of Geneva, 40 blvd. Du Pont d'Arve, 1211 Geneva 4, Switzerland
[d] Institute of Risk Analysis, Prediction and Management (Risks-X) Academy for Advanced Interdisciplinary Studies Southern University of Science and Technology (SUSTech), Shenzhen, China

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ABSTRACT

We theoretically model and empirically quantify the feedback effect of delta hedging for the spot market volatility of the forex market. We start from an economy with two types of traders, an aggregated option market maker (OMM) and an aggregated option market taker (OMT), whose exposures reflect the total outstanding positions of all option traders in the market. A different hedge ratio of the OMM and OMT leads to a net delta hedge activity, which introduces market friction. We represent this friction by a simple linear permanent impact model. This approach allows us to derive the dependence of the spot market volatility on the gamma exposure of the traders. Our theoretical model shows that the spot market volatility is increased (decreased) by a negative (positive) gamma exposure of the OMM, whereby the amount of the increase depends on the net delta hedge amount executed in the spot market. To empirically test this model, we first reconstruct the aggregated OMM's gamma exposure by using trade repository data and find that it is negative. The empirical analysis performed over the period from 21st October 2017 to 30th June 2018 using our reconstructed OMM data then strongly supports our theoretical model: The gamma exposure of the OMM turns out to be highly significant for the spot market volatility and, as expected, the spot market volatility is increased with the OMM's short gamma exposure. Quantitatively, a negative gamma exposure of the OMM of approximately -1000 billion USD (which is around what we observe from our reconstructed OMM data) leads to an absolute increase in volatility of 0.7% in EURUSD and 0.9% in USDJPY.

1. Introduction

This paper addresses the theoretical description and empirical quantification of the effect of delta hedging strategies on spot market volatility. By accounting for a linear, permanent market impact of spot market trades, we show that, in theory, spot market volatility is influenced by the net delta hedge demand arising from the options market. This theory is subsequently empirically validated with the use of trade repository data.

* Corresponding author.
E-mail addresses: Benjamin.Anderegg@snb.ch (B. Anderegg), fulmann@ethz.ch (F. Ulmann), dsornette@ethz.ch (D. Sornette).

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1.1. Literature review

Option pricing theory has been studied extensively since the seminal works of Black and Scholes (1973) and Merton (1973). Their theory is fundamentally based on an explicit trading strategy for the underlying asset and riskless bonds whose final payoff is equal to the payoff of the derivative security at maturity. As this trading strategy provides an insurance against the risk that comes with buying and selling the derivative, it is called a dynamic hedging strategy (dynamic, since the replication should ideally be continuous) Sircar and Papanicolaou (1998).

A crucial assumption of the Black–Scholes model is a frictionless and elastic market, i.e., a market in which trades have no price impact, neither temporary nor permanent Gueant and Pu (2015). However, a vast body of work concerned with market microstructure clearly shows that one can reject the assumption of a frictionless market Farmer et al. (2006). In the context of dynamic hedging strategies, this implies a feedback effect on market dynamic parameters such as the spot price and volatility structure: the dynamic hedging strategy requires the investor who sells (buys) an option to sell (buy) the asset in the spot market when its price declines and to buy (sell) it when its price goes up. From the nature of such a trading pattern, one can expect an increase (decrease) in spot market volatility Frey and Stremme (1997) Sircar and Papanicolaou (1998). Indeed, spot market volatility has been reported to have changed due to the introduction of futures and options Miller (1997) MacKenzie (2008). Note that this feedback effect can even result in two extreme price dynamics, market crashes and pinning. For example, Almgren (2014) attributes the Oct. 15, 2014, US Treasury price action, in which the US Treasury market underwent the largest intraday move since 2009, to the hedging of short option positions held by the dealer community. Avellaneda and Lipkin (2003), on the other hand, discuss the case where hedge funds, which have a long option position, pin the spot price to the strike at expiration due to their hedging behaviour. Furthermore, since changes in market dynamic parameters (e.g., spot market volatility) also change the derivative's price and therefore its dynamic hedging strategy, market dynamic parameters cannot be regarded as exogenous. In this context, Lions and Lasry (2006) cite practitioners who observe that the future values of the parameters they use for the pricing of options might be modified by their own activities (which is called a feedback effect).

Different work streams have emerged from the discussion of market frictions in the context of dynamic hedging strategies of options. Some are more focused on accounting for market frictions in the pricing of options (and in deriving an optimal hedging strategy), while others are more concerned by describing the actual feedback effect that dynamic hedging has on market dynamics. Our paper contributes to the latter stream. More specifically, we focus on the effect that delta hedging has on the spot market volatility. Nevertheless, we first present a brief literature review for both streams.

Accounting for feedback effects in the Black–Scholes pricing model has been the objective of a large body of academic literature. First, Brennan and Schwartz (1989) and Frey and Stremme (1997) focus on the derivation of the price dynamics through a clearing condition. Sircar and Papanicolaou (1998) even goes a bit further and introduces an option pricing model that accounts self-consistently for the effect of dynamic hedging on the volatility of the underlying asset. They find that spot market volatility increases by approximately 10%-18% for traders who write options and hedge their delta in the spot market.

The other approach is to account for the price impact of hedging trades in the spot market directly. In this approach, the price impact is often decomposed into temporary and permanent price impact components Almgren and Chriss (2000).

Temporary price impact considerations are mostly relevant in the context of optimal hedging strategies, as presented in Rogers and Singh (2010) and Naujokat and Westray (2011). The basic problem is that continuous hedging is prohibitive due to execution costs stemming from temporary market impact, whereas low-frequency hedging leads to large tracking error of the frictionless Black–Scholes delta hedge Gueant and Pu (2015).

The consideration of a permanent market impact is dedicated to the hedging of derivatives in situations where the notional of the option hedged is such that the delta-hedging is non-negligible compared to the average daily volume traded on the spot market Bouchard et al. (2017). Note that this large position may be the position of a single large trader or may be the aggregate position of a collection of traders, for example, the entire sell-side community, who all hold similar positions and who all hedge while their counter-parties do not Li and Almgren (2016). Permanent price impact is e.g. considered by Abergel and Loeper (2016),Loeper (2013) and Bouchard et al. (2016). All of their theoretical works reveal that the market dynamics of the spot market are changed by dynamic hedging of derivatives. Thereby Lions and Lasry (2006), also using a permanent market impact, find that the impact upon the volatility of the asset depends on the gamma of the option of the dominant single large investor they model. Further, Li and Almgren (2016) then take both temporary (with quadratic execution costs) and permanent price impacts into account (following Almgren and Chriss (2000)) and find that permanent market impact causes an increase or decrease in the realised volatility of the market price, depending on whether the large investor or the entire community of hedging traders is net long or short options.

1.2. Our contribution

We will focus on the theoretical and empirical discussion of the feedback effect stemming from delta hedging strategies in the FX spot market.

First, similar to the approach presented in Frey and Stremme (1997), we work in a model economy with only two agents. Our agents are represented by two types of traders, namely, an aggregated option market maker (OMM), who provides liquidity to the options market, and an aggregated option market taker (OMT), who consumes liquidity. As a consequence of this setup, the OMM and the OMT always have exact opposite positions from one another, and each of their books represent the total outstanding position on the options market.

We then assume that the two traders' hedging trades in the spot market have a linear, permanent market impact and no temporary market impact. This framework results in a simple volatility model that shows that the spot market volatility increases above its fundamental value (which is the volatility in case of no market impact of delta hedging trades) if the hedge ratio of the trader, who is net short options (and therefore has a negative gamma exposure) is larger than the hedge ratio of the trader who is net long options. Note that this model economy is motivated by the idea that typical OMMs, e.g., large banks, dynamically hedge their positions while OMTs, e.g., investors, do not. This would result in an asymmetric net delta hedge demand, which will lead to an increase (if the OMM is short options) or decrease (if the OMM is long options) in the spot market volatility. Thus, our model framework and our theoretical results are similar to those presented by Lions and Lasry (2006) and Li and Almgren (2016). Nevertheless, nowhere in the paper do we a priori assume that the OMM hedges a larger share of her exposure compared to the OMT.

Second, we want to validate this model empirically for the FX market. While much work has been done on the theoretical side, empirical work on feedback effects arising from dynamic hedging has been limited. This gap exists because an empirical study of such feedback effects requires proprietary data, e.g., data that allow for the identification of the hedging demand. Our paper bypasses the need for proprietary data by reconstructing it from trade-repository data (see below).

So to validate our model, we first need to obtain the data of the aggregated OMM and OMT. The BIS triennial survey 2016 states that the FX market turnover is composed of 33% spot transactions and 67% derivatives such as FX swaps, outright forwards and options BIS (2016). Since 2016, those 67% are subject to mandatory reporting in the US, for which our main data source in this analysis, the Depository Trust and Clearing Corporation (DTCC) trade repository, is an authorised trade repository covering approximately 20% of the worldwide options market transactions Shtauber and Marone (2014). These transaction data allow for profound insights on the market positioning in terms of new and outstanding positions. DTCC positioning data have been used previously to analyze spillover effects from the FX options market on the FX spot market but mostly with a focus on the spillover effect on the spot price rather than its volatility Weng and Grover (2017) Shtauber and Marone (2014) Winkler (2017). The downside of the DTCC data is that we do not know the side behind each trade, and thus we do not know whether the transaction is buy or sell initiated. However, this information is necessary to reconstruct the aggregated exposure of the OMT and OMM, as we assume that the OMM is always the liquidity provider and the OMT is always the liquidity taker. We solve this problem by using a simple trade classification algorithm (the quote rule of Hasbrouck (1988)) to classify the aggressor side behind each option trade stored in the DTCC repository. [1] Thus, we can identify the OMM (liquidity provider) and OMT (liquidity taker) behind each trade and can therefore also calculate their aggregated position at each point in time. This allows us to estimate our volatility model. [2]

The remainder of this paper is structured as follows. The second section introduces our volatility model. The third section briefly explains how we reconstruct the OMM's position behind each transaction of the DTCC trade repository data. In the fourth section, we show our empirical results, supporting our volatility model structure. In the fifth section, we discuss our results and draw conclusions from our findings.

2. A model for the feedback effects of delta hedging

2.1. Derivation of the volatility model

In this section, we derive an expression for the spot market volatility that includes the price impact of the delta hedging activity. First, we assume the observable exchange rate, denoted by S_t, is given by a linear stochastic differential equation.

$$dS_t = \mu_t S_t dt + \sigma_t S_t dW_t \tag{2.1}$$

Further we first assume that the non-observable fundamental value of the exchange rate F_t, which has the interpretation of the market price unaffected by delta hedging (the friction in our model), follows a geometric Brownian motion.

$$dF_t = \alpha F_t dt + \nu F_t dW \tag{2.2}$$

Applying Ito's lemma to the log-price changes $df_t = d(ln(F_t))$ results in $df_t = \frac{dF_t}{F_t} - \frac{1}{2}\nu^2 dt$. Using dF_t from Eq. (2.2) we can then write df_t as

$$df_t = \underbrace{\left(\alpha - \frac{1}{2}\nu^2\right)}_{\alpha'} dt + \nu dW = \alpha' dt + \nu dW \tag{2.3}$$

Now we assume that there are only two market participants, the OMM and the OMT. Their net delta exposures are $N_t^{OMM}\Delta_t^{OMM}$ and $N_t^{OMT}\Delta_t^{OMT}$, respectively, where N_t is the notional (always positive) and Δ_t is the option delta (the delta is positive for a

[1] See Frömmel et al. (2020) for an overview on trade classification algorithms and their quality.

[2] Note that we only examine vanilla options, as the details of exotic options in the DTCC trade repository are not sufficient.

long call and a short put and negative for a short call and a long put). Both hedge their overall delta exposure to a certain degree in the spot market, given by the hedge ratios h^{OMM} and h^{OMT}, which we assume to stay constant in time. Assume that the market participants hedge their risk at discrete intervals. If the price changes by a certain percentage between time t and $t+1$, the net traded delta hedge volume in the spot market at time $t+1$ is given by the negative change of the delta exposure between t and $t+1$, multiplied by the hedge ratio, so $\Delta V^{OMM}_{t+1} = h^{OMM} N^{OMM}_t \left(-\left(\Delta^{OMM}_{t+1} - \Delta^{OMM}_t\right)\right)$ for the OMM and $\Delta V^{OMT}_{t+1} = h^{OMT} N^{OMT}_t \left(-\left(\Delta^{OMT}_{t+1} - \Delta^{OMT}_t\right)\right)$ for the OMT. Note that in a market with two participants, the notional is the same for both participants ($N^{OMM}_t = N^{OMT}_t = N_t$), and the delta of the OMM is always equal to the negative delta of the OMT ($N_t \Delta^{OMM}_t = -N_t \Delta^{OMT}_t$). Furthermore, the positions of each trader represent the total outstanding position in the market (which is why our two agents are the OMM and the OMT, as by definition the aggregated positions from all OMMs sum up to the total outstanding positions in the market). Defining the asymmetry in the hedge ratio of the OMM and the OMT from the OMM's point of view as $h^{OMM}_{net} = h^{OMM} - h^{OMT}$, the total net traded delta hedge volume between t and $t+1$ is given by

$$\Delta V_{t+1} = \Delta V^{OMM}_{t+1} + \Delta V^{OMT}_{t+1} \tag{2.4a}$$

$$= h^{OMM} N^{OMM}_t \left(-\left(\Delta^{OMM}_{t+1} - \Delta^{OMM}_t\right)\right) + h^{OMT} N^{OMT}_t \left(-\left(\Delta^{OMT}_{t+1} - \Delta^{OMT}_t\right)\right) \tag{2.4b}$$

$$= \left(h^{OMM} - h^{OMT}\right) N_t \left(-\left(\Delta^{OMM}_{t+1} - \Delta^{OMM}_t\right)\right) \tag{2.4c}$$

$$= h^{OMM}_{net} N_t \left(-\left(\Delta^{OMM}_{t+1} - \Delta^{OMM}_t\right)\right) \tag{2.4d}$$

In the continuous limit, Eq. (2.4d) has the following form:

$$dV_t = -h^{OMM}_{net} N_t d\Delta^{OMM}_t \tag{2.5}$$

Now we assume that the traded delta hedge volume executed in the spot market, dV_t, impacts the spot market price in a linear and permanent fashion, with β defined as the log-price change (market impact) per traded volume. This is the friction that we introduce and which will, as we see later, result in the feedback effect discussed in this paper. Thus, the observed log price change at time t, ds_t, is given by the log price change of the fundamental price at time t, df_t, as well as the price impact of the delta hedge volume executed in the spot market, βdV_t. E***q. 2.6 reflects this ansatz.

$$ds_t = df_t + \beta dV_t \tag{2.6a}$$

$$= df_t - \beta h^{OMM}_{net} N_t d\Delta^{OMM}_t \tag{2.6b}$$

We thereby assume that the price impact β is a constant. This assumption basically reflects Kyle's price impact model (the price impact is permanent, linear and depends on the volume of the transaction only), presented in Kyle (1985). The assumption that the permanent price impact scales linearly with the transaction volume is also in line with current price impact theory Bouchaud (2010).

Applying Ito's lemma to the log-price changes $ds_t = d(ln(S_t))$ results in $ds_t = \frac{dS_t}{S_t} - \frac{1}{2}\sigma_t^2 dt$. Using this, as well as ds_t from Eq. (2.6b), we can then write dS_t as

$$dS_t = S_t df_t + \frac{1}{2}\sigma_t^2 S_t dt - \beta h^{OMM}_{net} N_t S_t d\Delta^{OMM}_t \tag{2.7}$$

Applying Ito's lemma to $d\Delta^{OMM}_t$ with respect to dS_t leads to

$$d\Delta^{OMM}_t = \underbrace{\left(\underbrace{\frac{\partial \Delta^{OMM}_t}{\partial t}}_{-Charm^{OMM}_t} + \mu_t S_t \underbrace{\frac{\partial \Delta^{OMM}_t}{\partial S}}_{\gamma^{OMM}_t} + \frac{1}{2}\sigma_t^2 S_t^2 \underbrace{\frac{\partial^2 \Delta^H_t}{\partial S^2}}_{Speed^{OMM}} \right)}_{\gamma'_t} dt + \sigma_t S_t \underbrace{\frac{\partial \Delta^{OMM}_t}{\partial S}}_{\gamma^{OMM}_t} dW_t \tag{2.8}$$

The option greeks are labeled with brackets. γ^{OMM}_t, which is important in our theory, is the option gamma of OMM at time t. Inserting 2.3 and 2.8 into 2.7 results in

$$dS_t = \left(\alpha' S_t + \frac{1}{2}\sigma_t^2 S_t - \beta h^{OMM}_{net} N_t S_t \gamma'_t\right) dt + \left(v S_t - \beta h^{OMM}_{net} N_t S_t^2 \sigma_t \gamma^{OMM}_t\right) dW_t \tag{2.9}$$

Comparing Eq. (2.9) with 2.1 results in

$$\sigma_t = \frac{v}{\left(1 + \beta h_{net}^{OMM} N_t S_t \gamma_t^{OMM}\right)} \tag{2.10}$$

2.2. Discussion of the volatility model

We quickly discuss the intuition behind our presented model now. $h_{net}^{OMM} S_t N_t \gamma_t^{OMM}$ is the change of the total market's hedged delta exposure in base currency in response to a relative change in S_t. It is equal to $h^{OMM} S_t N_t \gamma_t^{OMM} + h^{OMT} S_t N_t \gamma_t^{OMT} = h_{net}^{OMM} S_t N_t \gamma_t^{OMM}$ by using the asymmetry in the hedge ratio of the OMM and the OMT from the OMM's point of view ($h_{net}^{OMM} = h^{OMM} - h^{OMT}$) as well as the fact that $\gamma_t^{OMM} = -\gamma_t^{OMT}$. The negative of $h_{net}^{OMM} S_t N_t \gamma_t^{OMM}$ is then exactly the amount which has to be executed in the spot market in response to a relative change in S_t in order to keep the hedged delta exposure constant. And if there is price impact β, this execution influences the volatility v by the factor $\left(1 + \beta h_{net}^{OMM} S_t N_t \gamma_t^{OMM}\right)^{-1}$.

In order to simplify the notation, we now define the **market's delta-hedged gamma exposure** Γ_t^M

$$\Gamma_t^M = h_{net}^{OMM} S_t N_t \gamma_t^{OMM} \tag{2.11}$$

Using the market's delta-hedged gamma exposure, given by Eq. (2.11), we can then rewrite Eq. (2.10) as

$$\sigma_t = \frac{v}{\left(1 + \beta \Gamma_t^M\right)} \tag{2.12}$$

We see that the spot market volatility of a market unaffected by delta hedging, v, is rescaled by a factor $\left(1 + \beta \Gamma_t^M\right)^{-1}$. Thus it is the Γ_t^M that determines whether the volatility is scaled up or down. If Γ_t^M is positive, the total market's hedged delta exposure increases in response to an increase in S_t. The negative of this increase in the delta exposure is then executed in the spot market, i.e. the market is selling the underlying if S_t increases, which counteracts the increase in S_t (if there is a price impact) and thus lowers the fundamental volatility v. But if Γ_t^M is negative, the total market's hedged delta exposure decreases in response to an increase in S_t. The negative of this decrease in the delta exposure is then executed in the spot market, i.e. the market is buying the underlying if S_t increases, which supports the increase in S_t (if there is a price impact) and thus increases the fundamental volatility v.

So the intuition behind the above described dynamic stems from the fact that a short delta-hedged gamma exposure of the market is associated with a short position in the net hedged option contracts of the market, which then again implies a net delta hedging strategy of the market in which she has to buy the asset if the spot price increases and sell the asset if the spot price decreases. This trading behavior leads to an increase in the spot market volatility given that the hedging trades are assumed to have a price impact (vice versa for the long delta-hedged gamma exposure of the market).

Whether Γ_t^M is positive or negative thereby depends on the gamma of the OMM, γ_t^{OMM} (or the gamma of the OMT as together they represent the full market), as well as on the asymmetry in the hedge ratio of the OMM and the OMT, $h_{net}^{OMM} = h^{OMM} - h^{OMT}$. In other words, to gauge the feedback of the option market on the spot market volatility, the gamma exposure of the trader who hedges a larger part of her delta exposure compared to the other trader is relevant. As described in Eq. (2.11), we call this the market's delta-hedged gamma exposure. And while the direction of the feedback effects depends on the sign of Γ_t^M, the strength of the feedback effect depends on the absolute value of the market's delta-hedged gamma exposure $|\Gamma_t^M|$ and on the price impact β. Note that as the volatility cannot be below zero by definition, the factor $\beta\Gamma_t^M$ must be in the open interval $(-1, +\infty)$.

Fig. 1 displays the above discussed model dynamics and Fig. 2 presents a case study for the case where the OMM holds a short call option and $h_{net}^{OMM} = 1$.

3. Reconstruction of the OMM's gamma exposure from trade repository data

In order to empirically validate Eq. (2.10), we need data on the gamma exposure of the OMM, $S_t N_t \gamma_t^{OMM}$ (or, equivalently, as we only have two traders in our model, the gamma exposure of the OMT, $S_t N_t \gamma_t^{OMT}$). In this section we discuss how we construct data on the gamma exposure of the OMM, $S_t N_t \gamma_t^{OMM}$. We thereby focus on the FX options market.

The Dodd-Frank Act, which was signed into law in July 2010, requires the reporting of all over-the-counter FX option transactions that were traded in the US, or involve US persons, to a data repository. Note that there are several data repositories that market participants can report to. We use the Depository Trust and Clearing Corporations repository (DTCC), which covers approximately 20% of the worldwide options market transactions Shtauber and Marone (2014).

Option trades stored in the DTCC trade repository are the result of OTC transactions. Large banks mostly act as liquidity providers for their clients, e.g., hedge funds, corporations, institutional investors, asset managers or private clients, which then act as liquidity takers. The banks provide liquidity by constantly offering a bid and ask price (which is quoted in terms

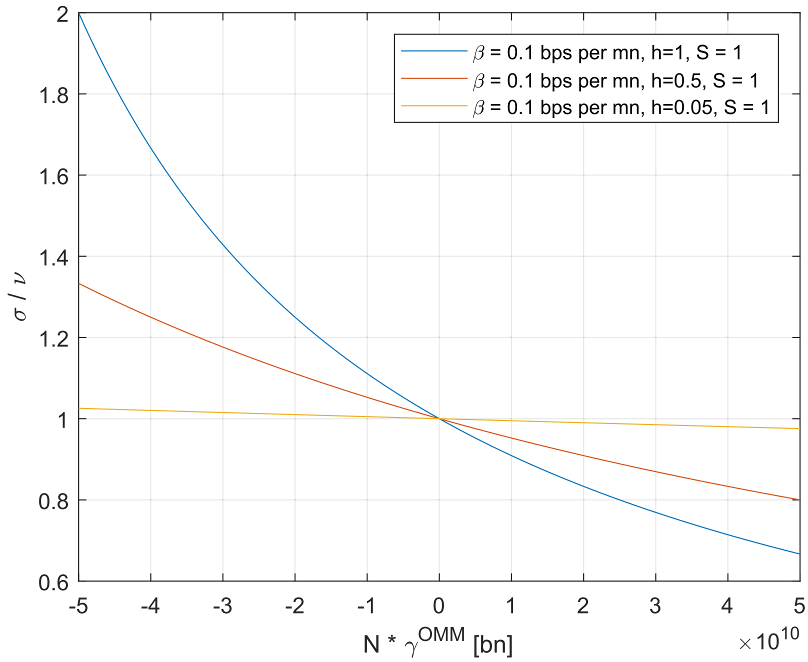

Fig. 1. Discussion of our volatility model: This figure shows the dependence of $\frac{\sigma_t}{\nu}$ as a function of $N_t\gamma_t^{OMM}$ as predicted by Eq. 2.10. On the vertical axis, we display the observable spot market volatility normalized by the volatility of a market unaffected by delta hedging, so $\frac{\sigma_t}{\nu}$. On the horizontal axis, we display the factor Γ_t^M, where we fix the market impact to $\beta = 0.1[bps/mn\ \ base\ \ currency]$, the exchange rate to $S_t = 1$ and h_{net}^{OMM} to three different values. We thereby always assume that the OMM hedges a larger part of its delta exposure, so $h_{net}^{OMM} > 0$, and thus the sign of her gamma exposure is the relevant one for the discussed feedback effect. Thus Γ_t^M reduces to $N_t\gamma_t^{OMM}$, as all other variables are fixed. We see that the volatility is scaled up or down depending on the sign of the delta-hedged gamma exposure of the market Γ_t^M (or rather by the sign of the gamma exposure of the OMM, as we assumed $h_{net}^{OMM} > 0$). The strength of this effect thereby depends non-linearly on the absolute value of the gamma exposure $N_t\gamma_t^{OMM}$.

of implied volatility) to their clients, and the clients then take the liquidity by either buying on the ask price, this deal is called a paid, or selling on the bid price, this deal is called a given. The trade is then reported to the DTCC trade repository, but the information on whether the deal happened on the bid or the ask price is not included.

As presented before, our model economy is motivated by this structure of the OTC option market, and models only two traders, the option market maker (OMM) and the option market taker (OMT). The OMM represents the aggregate of the liquidity providers (mostly banks) and the OMT represents the aggregate of the liquidity takers (hedge funds, corporations, institutional investors, asset managers and private clients). All deals are attributable to the aggregated position of the OMT, whereas the OMM then just has the exact opposite position.

The main problem is that the public DTCC data does not provide information on the type of deal that happened, so whether a deal was a paid or a given. Thus, we need to reconstruct this information in order to quantify the OMT's positioning in the options market. We thus follow a trade classification procedure. In the first step, the implied volatility of each option transaction stored in the DTCC trade repository is reconstructed by numerically minimizing the difference between the premium that was paid (which is reported to the DTCC trade repository) and the premium that Black's model suggests for the trade (based on the specifications that are reported to the DTCC trade repository). [3] Next, this reconstructed implied volatility, which was traded at a certain time (that is also reported to the DTCC trade repository), is matched onto the simultaneous reference bid-ask price stream for that time, that we take from Bloomberg. Following the quote rule of Hasbrouck (1988), we label the transaction a paid (buy order on the ask price) if the reconstructed implied volatility is closer to the ask price compared to the bid price or a given (sell order on the bid price) if the reconstructed implied volatility is closer to the bid price compared to the ask price. Those deals then reflect the OMT's position. And as the OMM is always on the other side of each deal, it follows that she has a short position if the transaction is labeled a paid and a long position if the transaction is labeled a given. A discussion of the quality of our reconstruction proceedure is presented in B.3.

As we now know the OMM's position behind each transaction, we can calculate the delta and gamma exposure of the OMM at any given point in time. We simply calculate the delta and gamma of the OMM's open positions at any given point in time, multiply it by the signed notional of the corresponding position (i.e., the negative notional if the OMM is short the option) and take the corresponding sum over all open positions. This results in the aggregated delta and gamma exposure of the OMM's portfolio at any given point in time ($S_tN_t\gamma_t^{OMM}$), which we now can use to validate our volatility model.

[3] For other asset classes, other option pricing models might be appropriate.

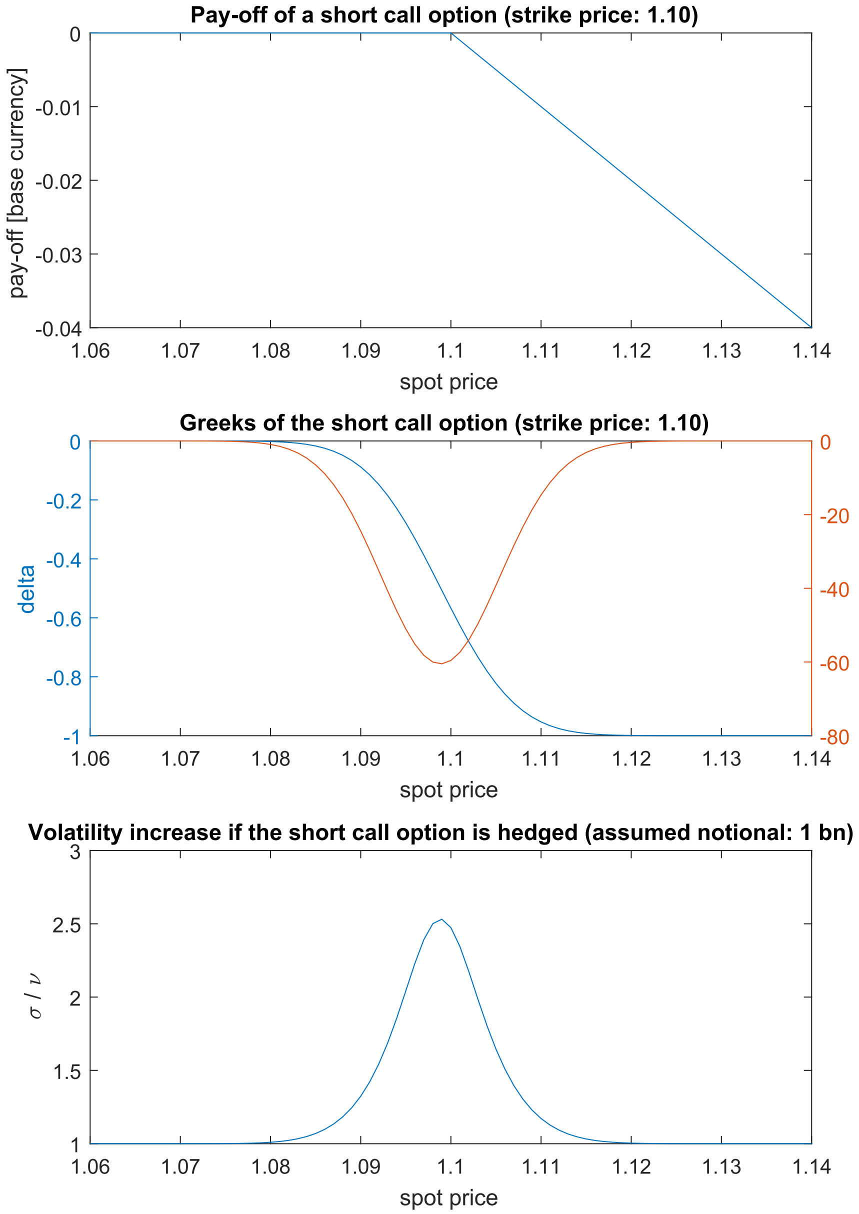

Fig. 2. Case study of our volatility model: We assume that the OMM has a short call position. The upper panel shows the pay-off at maturity of the OMM's short call position, with a strike price at $K = 1.10$. The middle panel displays the corresponding delta and gamma profile of the short call option with a volatility of 6% and a time-to-expiry of three months. We thereby see that the delta is decreasing in the spot price, and thus the corresponding delta hedge profile (which is the negative of the delta profile) is increasing in the spot price. This implies that the OMM needs to buy the underlying if the spot price goes up and sell the underlying if the spot price goes down. The corresponding gamma profile is negative. Now we assume that the notional behind the short call option is one billion ($N_t = -1$bn) and that the OMM hedges her total position in the spot market, while the OMT does not hedge her position at all (hedge ratio of $h_{net}^{OMM} = 1$). Note that thus the OMM's gamma position is equal to the market's delta-hedged gamma position. The resulting increase in the volatility ($\frac{\sigma}{\nu}$, based on Eq. 2.10 with a permanent market impact of $\beta = 0.1[bps/mn]$, a hedge ratio of $h_{net}^{OMM} = 1$ and a notional of $N_t = -1$bn) is displayed in the lower panel. We see that the volatility locally increases at spot prices for which the gamma profile of the .OMM is negative.

A more detailed discussion on the underlying DTCC data quality and on how the OMM's positioning is derived from the DTCC data can be found in the appendix.

3.1. Discussion of the OMM's gamma exposure reconstructed from DTCC data

For the reconstruction of the OMM's gamma exposure from DTCC data we focused on the FX pairs EURUSD and USDJPY, as for those two FX pairs the outstanding volume in the FX options market is large enough to expect significant feedback effects on the FX spot market. As described before, we reconstructed the delta and gamma exposure of the OMM at all points in time. The corresponding results are displayed in Fig. 3, where each trading day is split into 12 two-hour intervals. [4]

As we see in Fig. 3, the OMM is short gamma at all times. This implies that the OMM is shorting put and call options (on a net basis). This is in line with our expectation. First, most investors are purchasers of options rather than sellers. They use options as a hedge or with consideration of the spot price and, therefore, mostly purchase options. Furthermore, they refrain from going short an option contract (put or call), as the risk involved in doing so is too high. Those traders most certainly trade as market takers in the market, e.g., as clients of banks. On the other side of each transaction of a market taker sits a market maker, e.g., the banks; consequently, the market maker is expected to have a short option position and therefore a short gamma position. Second, a short position in an option implies a short vega position, and as the implied volatility usually trades with a premium to the realized volatility, this is one way for option market makers to make money. For those two reasons, the OMM is expected to have a net short gamma position, which is exactly what we observe. As stated above, we see from our data that the OMM is short 75% of all traded option contracts in EURUSD and 78% of all traded options in USDJPY.

For the delta exposure, we have no a priori expectation of how it should look, since the delta exposure is not directly related to the positioning (if the OMM is long or short options) but rather is a function of where the spot price trades relative to the notional weighted strike price of the aggregated OMM's positions.

If the OMM would hedge her total position in the spot market, her delta hedge strategy would be to trade the negative difference of her delta exposure between two points in time in the spot market. The OMM would then be delta neutral by the end of each time period. Note that, as gamma is the change in delta per unit change in the spot price (derivative of the delta with respect to the spot price), gamma indicates how much delta would have to be hedged if the spot price moved one unit. In that sense, gamma is a proxy for the intensity of the delta hedge activity.

4. Validation of the volatility model

As outlined in the previous sections, the market's delta hedged gamma exposure is assumed to generate feedback effects: a positive gamma exposure implies that the market sells the underlying asset in a rising market to hedge its delta exposure, and as the executed hedging trades in the spot market have a market impact, this activity should have a negative effect on the realised spot market volatility. Following the same line of argument, a negative gamma exposure should have a positive effect on the realised volatility. Our volatility model reflects this logic. In this section, we will validate and quantify this feedback effect empirically with the data on the gamma exposure of the OMM that we reconstructed from the DTCC data.

$$\sigma_t = \frac{\nu}{\left(1 + \beta h_{net}^{OMM} N_t S_t \gamma_t^{OMM}\right)} \tag{4.1}$$

We validate the volatility model presented in Eq. (4.1) with the use of MATLAB's nonlinear regression model Matlab,(2019). Thereby the data input is σ_t, the realised spot market volatility (annualised) for the time period between t and $t+1$, and $N_t S_t \gamma_t^M$, the reconstructed OMM's gamma exposure at time t. The parameters that we estimate in the regression model are ν and $\beta * h_{net}^{OMM}$ [5]. As the volatility strongly depends on the intraday liquidity cycle, we validate the model with daily data.[6]

Table 1 and Fig. (4) present the results of the nonlinear regression model.

The high values for the F-test strongly support our proposed model and thus the existence of the described feedback effect. And we see highly significant parameter estimates for the factor βh_{net}^{OMM} as well as for ν, which tells us that the gamma exposure of the OMM has high relevance for the spot market volatility.

The root mean squared error (RMSE) of the estimation is fairly low, given the high noise of daily volatility estimates and the fact that we analyse the observed volatility only with respect to one isolated effect (the feedback effect) in our model.

E.1 presents the analysis on intraday data in a random intercept linear mixed-effect model, for which we realise very similar parameter estimates and a high goodness of fit in terms of R^2 (a measure not valid for the here presented nonlinear model fit).

The results are very robust against changes in the estimation period, i.e. the significance or parameter estimate change with negligible amplitude upon a change in the estimation period.

The estimate of ν reflects the fundamental volatility, which can be interpreted as the spot market volatility in the absence of any delta hedging activity. It is higher for EURUSD than for USDJPY. This difference stems from the fact that the volatility data for the daily estimate for USDJPY originate from the Asian trading hours, in which the volatility is usually lower than during the European or American trading hours.

[4] UTC 00:00–02:00, 02:00–04:00, 04:00–06:00, 06:00–08:00, 08:00–10:00, 10:00–12:00, 12:00–14:00, 14:00–16:00, 16:00–18:00, 18:00–20:00, 20:00–22:00, 22:00–24:00

[5] Starting values for βh_{net}^{OMM} and ν are chosen as 0 and the convergence criteria is set at a tolerance of $1e-8$.

[6] For EURUSD, we look at the European trading session (UTC 09:00 to UTC 16:00), and for USDJPY, we look at the Asian trading session (UTC 2:00 to UTC 9:00). So we e.g. regress the gamma position of the OMM in EURUSD at 09:00 UTC onto the realized spot volatility between 09:00 and 16:00.

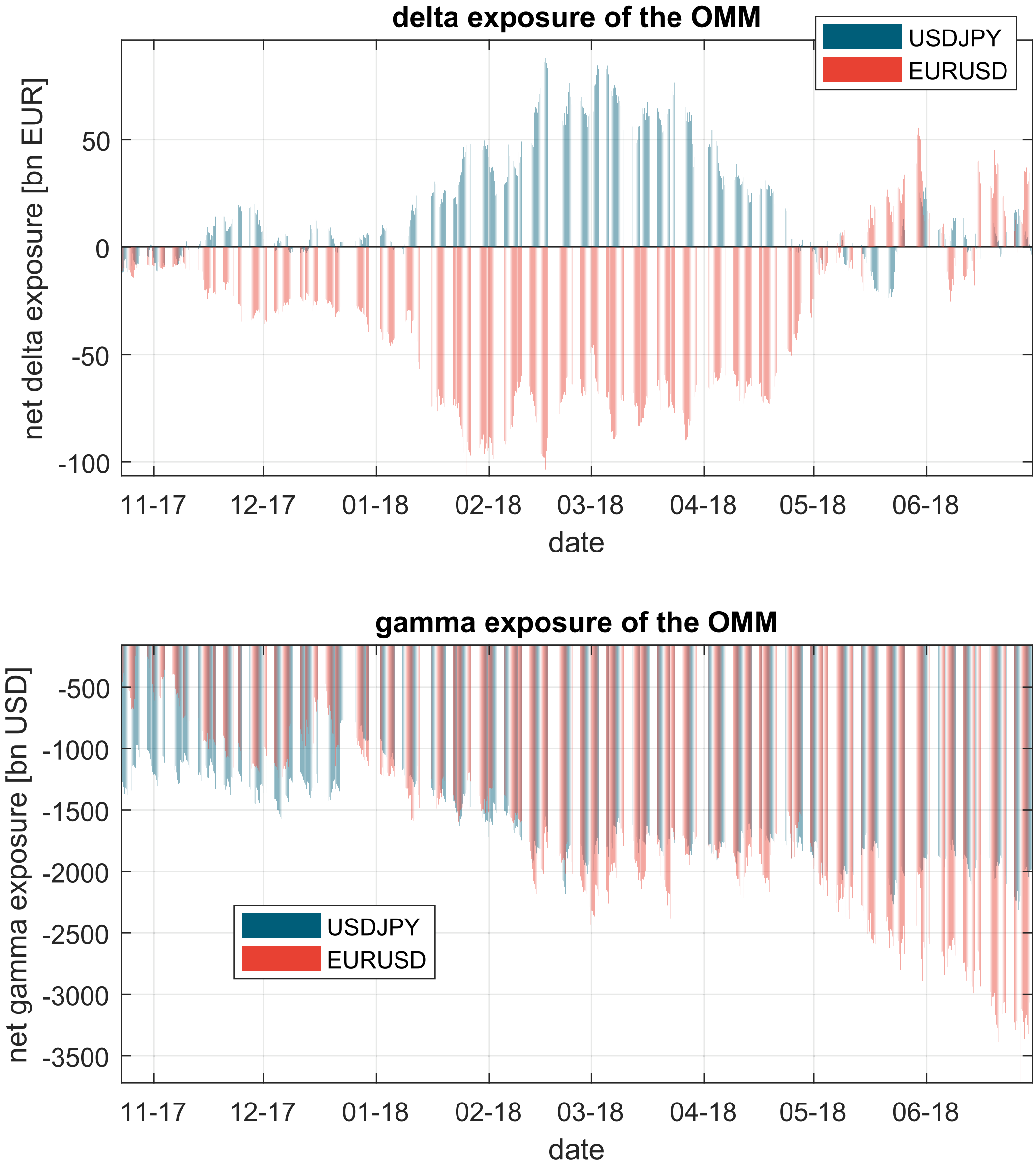

Fig. 3. Reconstructed delta and gamma exposure of the OMM: The figure shows the evolution of the accumulated net delta and gamma exposure of the OMM over the analysed time frame for EURUSD and USDJPY.

Table 1
This table shows the least-square fit results of the nonlinear model fit for EURUSD and USDJPY. t-statistics are presented in brackets. The stars *** denote significance at the 1% level.

	EURUSD	USDJPY
ν (%)	7.2***	5***
βh_{net}^{OMM} ($\frac{\%}{bn}$)	0.00009***	0.00015***
RMSE (min. σ, max σ)	2.1 (3.8, 17.5)	1.7 (3.0, 14.6)
F-test vs. const. model	1430***	1280***

The value of βh_{net}^{OMM} is not further disentangled here, as we would need an estimate of one of them, which is empirically difficult. But as the price impact β for sure is positive, we can conclude from the parameter estimates that h_{net}^{OMM} is also positive. This confirms that the OMM hedges a larger share of her delta exposure compared to the OMT, which is in line with our

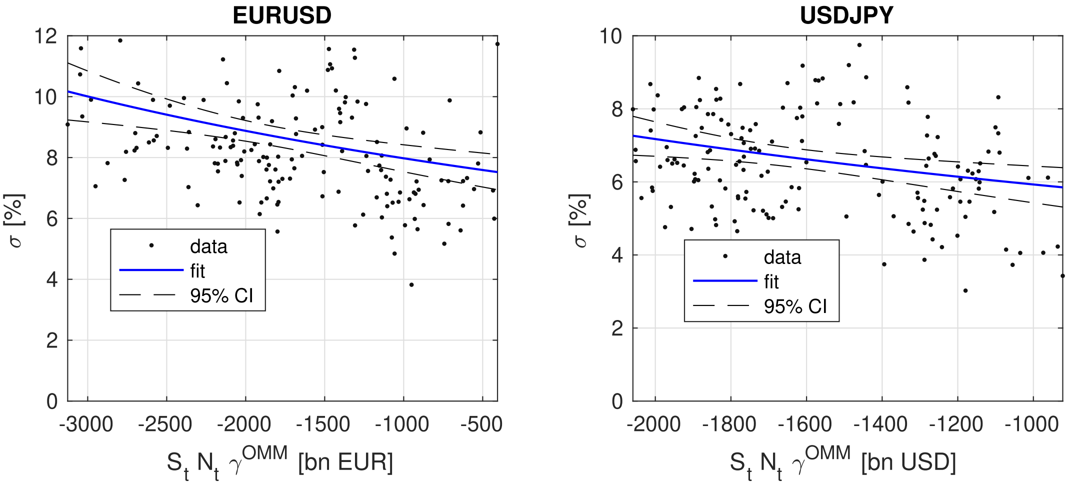

Fig. 4. In these panels, we show the nonlinear model fit for EURUSD and USDJPY (solid blue line) and the respective 95% confidence intervals (dashed black lines).

expectation that the OMM hedges his positions as he provides liquidity to the market as a service and thus needs to hedge his exposures.

The estimate of the parameter βh_{net}^{OMM} can now be used to quantify the feedback effect of the OMM's reconstructed gamma exposure on the observed spot market volatility. This discussion is presented in Fig. 5, using the fitted parameter values. In the upper panel, we present the fitted volatility model. The middle panel presents the absolute volatility increase due to the feedback effect, and the lower figure presents the relative volatility increase. We see that a negative gamma exposure of the OMM of approximately -1000 billion in the base currency (which is around what we observe in Fig. 3) leads to an increase of the EURUSD volatility above the fundamental volatility level of 0.7% [7] and to an increase of the USDJPY volatility above the fundamental volatility level of 0.9%. In the lower panel, we subsequently see that this gamma exposure of approximately -1000 billion in the base currency amounts to a relative increase above the fundamental volatility level of approximately 10% for EURUSD [8] and 18% for USDJPY. [9] This feedback effect stems from the fact that a short gamma exposure of the OMM (who is found to hedge a larger share of her delta exposure than the OMT) is associated with a short option position, which implies a delta hedging strategy in which the OMM has to buy the base currency if the spot price increases and sell the base currency if the spot price decreases. This leads to an increase in the spot market volatility given that the hedging trades have a price impact (vice versa for the long gamma exposure). [10]

Apparently, the feedback effect is stronger (per billion gamma exposure of the OMM) in USDJPY than in EURUSD. As the net hedge ratio h_{net}^{OMM} is probably of similar size in both currency markets, we can conclude that this higher effect can be attributed to a higher β in the USDJPY market (as then the higher estimate of βh_{net}^{OMM} in USDJPY can only be explained by a higher β). This can be explained by the higher liquidity of the EURUSD spot market and thus the lower market share of the delta hedging strategy in EURUSD and the lower price impact of the traded volume BIS (2016).

Lastly, note that very similar results were found with other empirical approaches, as e.g. with a random intercept linear mixed-effect model in which we used intraday data but accounted for the corresponding intraday time with a grouping variable. The intraday results are presented in the appendix E.

5. Discussion and Conclusion

To model the feedback effect of delta hedging for spot market volatility, we have proposed a model economy of two types of traders, an OMM and an OMT, whose exposures each reflect the total outstanding positions in the market. If the OMM and the OMT have different hedge ratios, this will result in a net delta hedge activity that introduces market friction and feedback effects. We conceptualise this friction by assuming a simple linear, permanent impact model for the net delta hedge volumes that are executed in the spot market. From this framework we subsequently derive a spot market volatility model, given by

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[7] $\sigma - \nu = \frac{\nu}{1-\beta * \ h_{net}^{OMM} \ * \ N_t S_t \gamma_t^{OMM}} - \nu = \frac{7.2\%}{1-0.00009\frac{\%}{bn} \ * \ 1000bn} - 7.2\% = 0.7\%$

[8] $\frac{\sigma-\nu}{\nu} = \frac{0.7\%}{7.2\%} = 10\%$

[9] Note that the true effect per billion is probably a bit smaller when we adjust for the fact that our OMM positioning is only based on the DTCC data, which represent only approximately 20% of the total options market Weng and Grover (2017). But as most relevant OMM report to DTCC, we regard it as a good representation of the total options market.

[10] Even though in theory one could conclude that our model is tradable, this is not the case, as no risk-less profit could be made based on our model if one takes transaction costs into account. A short discussion of that is presented in appendix D.

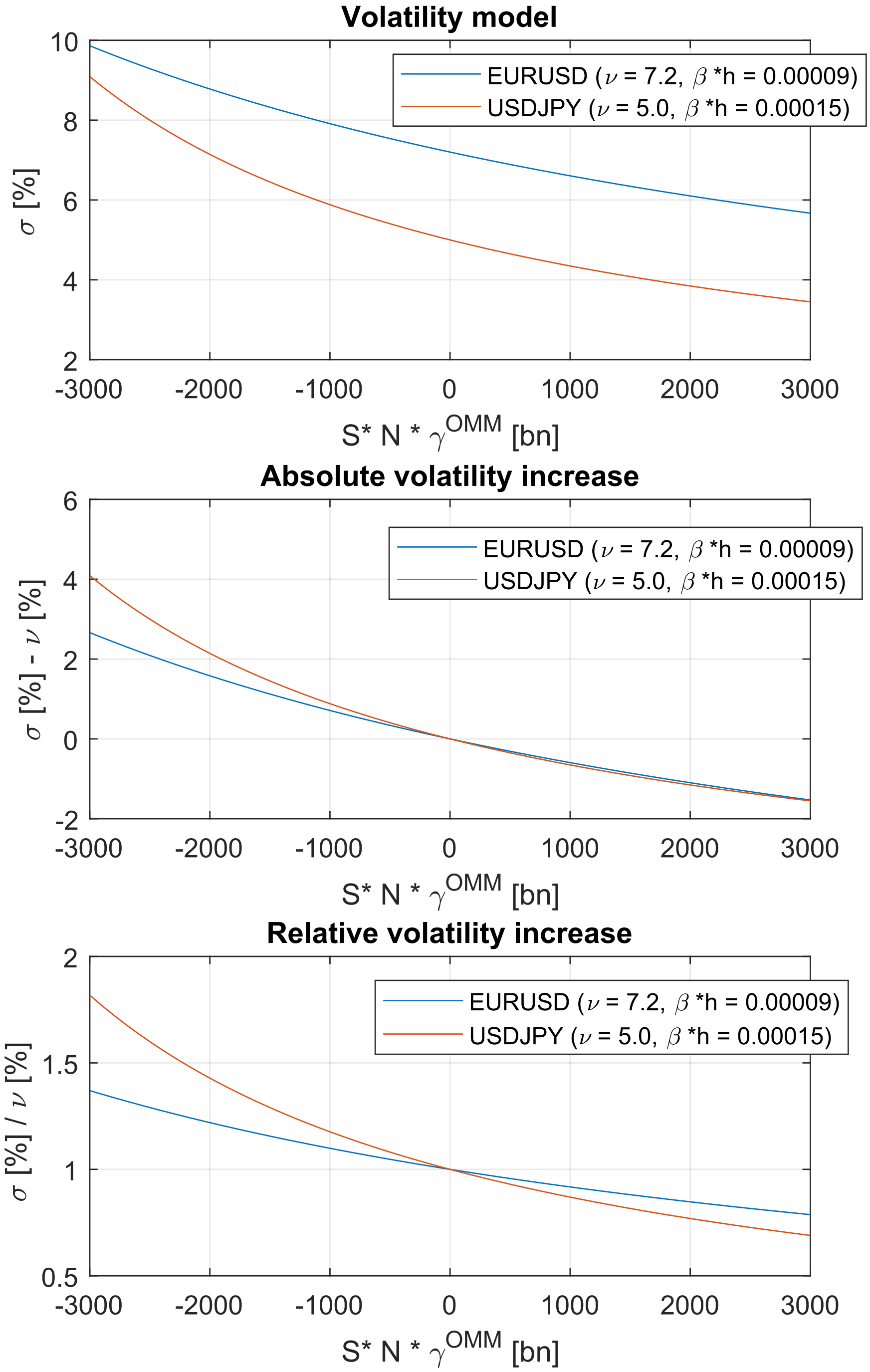

Fig. 5. In these panels, we discuss our volatility model with the fitted parameters for EURUSD and USDJPY. We present the fitted volatility model in the upper panel, the absolute volatility increase due to delta hedging activity in the middle panel and the relative volatility increase in the lower panel.

Eq. (2.12), that depends on the markets delta hedged gamma exposure as well as on the magnitude of the price friction (i.e. the price impact of delta hedge transactions in the spot market). For example, if the OMM has a negative gamma exposure and she hedges a higher degree of her delta exposure compared to the OMT (two of our empirical findings), the spot market volatility is structurally high, as the net delta hedge amounts, executed in the sport market, would be positive if the spot price moves up and negative if the spot price moves down (which results in an increase in the spot market volatility if the transactions come with a price impact). The implications of this model are in line with the theoretical results of Lions and Lasry (2006) and Li and Almgren (2016).

Feedback effects of this nature have not been empirically discussed, as the data to do so were not available. In the case of our volatility model, we needed the aggregated OMM's gamma exposure in order to empirically validate the model and therefore quantify the feedback effect. We reconstructed these data by using the publicly available DTCC trade repository data. This reconstructed net gamma exposure of the OMM was found to be negative, as expected: investors usually buy options with either consideration of the spot price or with the desire to hedge other positions. As those traders usually act as OMTs in the markets, we expect the OMT to be net long on options. As the OMM provides liquidity as a service to the market, his position will be the reverse of the OMT. Therefore, a net short position makes sense from our point of view.

We then use this reconstructed gamma exposure of the OMM to validate our volatility model. A least-square fit thereby shows a high goodness of fit, which strongly supports our model and the discussed feedback effect. The two main empirical findings are first, that the OMM hedges a larger share of her delta exposure compared to the OMT and second, the validation and quantification of the feedback effect. We will elaborate on the to findings here:

(1) We find that the OMM hedges a larger share of her delta exposure compared to the OMM. This, again, is in line with our expectation and in accordance with the line of reasoning presented above: As OMTs are usually investors who buy options with consideration of the spot price or to hedge other positions in their portfolio, they are expected to not hedge their option's delta. As the OMMs provide liquidity to the market as a service but usually without taking a view on the spot, they are assumed to hedge their large portfolio in order to reduce the risk involved in their business model. Therefore, we expected the hedge ratio of the OMM to be higher than that of the OMT, which is exactly what we found. Consequently, we also observe that the volatility is increased by the OMM's short gamma exposure.

(2) We find very strong statistical support of our volatility model structure and are able to use the model fit to quantify the feedback effect of the OMM's gamma exposure on the spot market volatility. A negative gamma exposure of the OMM of approximately -1000 billion in the base currency (which is around what we observe from our reconstructed OMM data) leads to an absolute increase in volatility of 0.7% in EURUSD and 0.9% in USDJPY. If we assume that the hedge ratios in both markets are the same, this difference can be directly explained by the higher market impact of a transaction in the USDJPY spot market compared to the EURUSD spot market, which makes sense, since the liquidity of the EURUSD spot market is higher than that of the USDJPY spot market. This absolute increase amounts to a relative increase in the volatility above its fundamental level (volatility level in the absence of any delta hedging) of approximately 10% for EURUSD ($\frac{0.7\%}{7.2}$) and 18% for USDJPY ($\frac{0.9\%}{5.0}$). Therefore, our results are in line with previous theoretical work on the feedback effect that delta hedging strategies have on the spot market volatility, as e.g., Sircar and Papanicolaou (1998) suggest a relative increase in the spot market volatility of approximately 18%.

Note that it would be wrong to conclude now that the options market introduces a destabilizing or stabilizing effect on the spot market via the delta hedging strategies per se. Whether the effect of the option market on the spot market is stabilizing or not only depends on the net gamma position of the OMM (as the OMM is found to hedge a higher share of her delta exposure compared to the OMT). However, since the OMM is usually short gamma, as other market participants refrain from selling options, the option market is found to lead to an increase in the spot market volatility. Moreover, while these effects appear to exist, implied volatility is still trading at a premium to the realized volatility! One would assume that the OMM would thus only stop selling options if her hedging strategy would drive the realized volatility above the implied volatility.

In future work, we intend to discuss how the persistence of the gamma exposure of the OMM could explain part of the autocorrelation that is observed in the spot market volatility; we also intend to discuss the local volatility phenomena and to examine the connection of an extreme gamma position in the context of market crashes.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Appendix A. Data Specification

A.1. Depository Trust and Clearing Corporations repository (DTCC) data

The Dodd-Frank Act, which was signed into law in 2010, requires the reporting of all over-the-counter FX option transactions that were traded in the US or involve US persons to a data repository. There are several data repositories to which market participants can report. We use the largest one of those, the Depository Trust and Clearing Corporations repository (DTCC) DTCC (2011).

The DTCC data are a representative source of the total options market. The share of the DTCC transactions among the total turnover in the options market, as published by the BIS, is approximately 20%. Due to the large volumes traded in the EUR-USD and USDJPY options market, we focus on those two instruments.

Most relevant information of the options trade is included in the repository, such as the type of options traded, the notional amount (volume), the strike price, the expiry date, the premium paid, the currency for quoting the notional and the premium as well as the option trade timestamp. The missing information is the investor type (hedge fund, corporation, asset manager, private client, etc.) and the side of the trade (if it was a paid or given); in addition, trades done outside the US are not represented. Our reconstruction logic fills one of the gaps, namely, the missing information about the side behind a transaction, by reconstructing the implied volatility of each trade and subsequently matching it with the reference bid/ask price at the time of the trade to check if the trade happened on the bid or the ask side (for more detail, see Appendices B.1 and B.2).

The key in this reconstruction and matching procedure therefore is to have accurate time stamps from the DTCC repository in a sense that they need to be reported immediately to the repository. To get an indication if this immediate reporting is given, we looked at the intraday turnover distribution of the DTCC data time stamps in order to check whether the distribution of the time stamps follows the usual intraday turnover pattern observed in all financial markets, or if there is an unusual aggregation of timestamps in the end of the day (due to delayed reporting). Fig. A.6 shows this analysis.

All of the above factors make the DTCC repository an ideal source for our analysis purposes.

A.2. General remarks on the data used in this paper

- The options price stream onto which the reconstructed implied volatilities are matched, comes from Bloomberg. Those options price streams are quoted in terms of implied volatility for standardised contracts (standard tenor and delta).
- The forward points and interest rates which we further used in the estimation of the implied volatility behind each option trade, as well as for the calculation of the delta and gamma of the OMM at each point in time, are also from Bloomberg.
- Only vanilla option contracts with a time to expiry below 600 days are taken into account.
- The observed annualised spot market volatility is calculated based on mid-price data from the corresponding primary FX market of the currency pairs (EBS).
- Analysis date range: 20.9.2017 to 30.6.2018.
- Intraday Bloomberg price streams are only available 60 days into the past. We thus cannot estimate the OMM's positions (from the DTCC data) for the period before the analysis period. Therefore, the estimated gamma exposure of the OMM is inaccurate in the beginning of the analysis period, as only the "new" transactions contribute to our estimate of the OMM exposure. However, the farther we go into the analysis period, the more accurately we can know the OMM exposure. As the median time to expiry of the analysed option contracts turned out to be 30 days, we leave out the first 30 days in our analysis to obtain a representative data set of the OMM positioning. All figures are displayed accordingly.

Appendix B. Derivation of the OMM's positioning behind each transaction

The DTCC repository provides useful information on each transaction but lacks one key point for our analysis: whether the deal was a paid (buyer initiated) or given (seller initiated). We solve this question by using a trade classification algorithm. First, we reconstruct the implied volatility of each trade by using the data from the DTCC trade repository. Second, using a trade classification algorithm, we match the reconstructed implied volatility onto a reference bid/ask price stream from Bloomberg and estimate whether it was traded on the bid or the ask price. By doing so, we estimate the side behind each transaction, that is, if it was a paid (active buy transaction) or a given (active sell transaction). Third, we derive the OMM's position, as she is always liquidity provider by definition (i.e., if the transaction was identified as paid, we assume the OMM is short the contract, as she provided liquidity on the ask).

B.1. Reconstruct the implied volatility from the transaction data

Here, we outline Blacks model. These formulas are used in the determination of the implied volatility.

$$C(F,\tau) = D \cdot (\Phi(d_+)F - \Phi(d_-)K) \tag{B.1}$$

$$P(F,\tau) = D \cdot (\Phi(-d_-)K - \Phi(-d_+)F) \tag{B.2}$$

$$d_\pm = \frac{1}{\sigma\sqrt{\tau}} \cdot \left(ln\left(\frac{F}{K}\right) \pm \frac{1}{2} \cdot \sigma^2 \tau \right) \tag{B.3}$$

$$d_\pm = d_\mp \pm \sigma\sqrt{\tau} \tag{B.4}$$

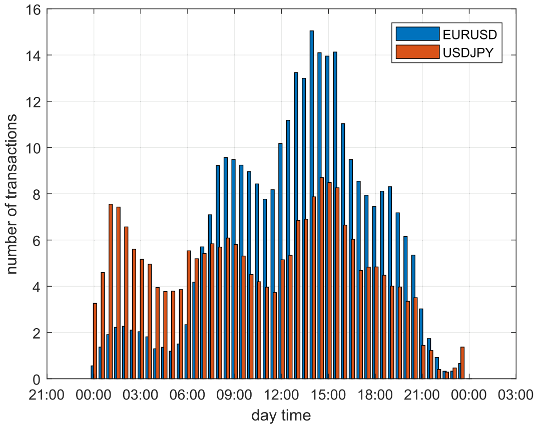

Fig. A.6. This figure shows the intraday turnover of the data that underlie our analysis. The intraday turnover, measured as the number of transactions per half hour, seems to follow a typical financial market turnover pattern and therefore serves as a strong indication that the trades are reported in a timely manner to the DTCC trade repository. This is a crucial condition for the analysis, as it means that the timestamps are reliable, which is a necessary precondition for the reconstruction of the side (if it was traded on the bid or the ask price). Furthermore, the volumes traded for EURUSD and USDJPY are comparable in size.

$$\tau = T - t \tag{B.5}$$

$$D = e^{-r_{priceccy} \cdot \tau} \tag{B.6}$$

$$F = S \cdot e^{\left(r_{priceccy} - r_{baseccy}\right) \cdot \tau} \tag{B.7}$$

Now we briefly explain the variables and comment on where the information for each variable comes from.

- Φ: Cumulative distribution function of the standard normal distribution
- T: Expiry date; from DTCC data
- t: Trade time stamp; from DTCC data
- τ: Time-to-expiry (also called tenor); can be calculated with Eq. (B.5)
- K: Strike price; from DTCC data
- $r_{priceccy}$: Interest rate of the price currency for the tenor τ; from Bloomberg; obtained by either looking at the bond yield of the corresponding country with tenor τ or, if τ is in between two standard bond contracts, calculating an interpolated yield from the two corresponding bond contracts (e.g., for a tenor of $\tau = 1.3$ years, the bond yield of the $\tau = 1$ year and $\tau = 2$ years is interpolated)
- $r_{baseccy}$: Interest rate of the base currency for the tenor τ; from Bloomberg; obtained by either looking at the bond yield of the corresponding country with tenor τ or, if τ is in between two standard bond contracts, calculating an interpolated yield from the two corresponding bond contracts.
- $F(\tau)$: Forward price (amount of price currency per unit of base currency in time τ); comes from Bloomberg; obtained by either looking at the forward points of the corresponding currency with tenor τ or, if τ is in between two standard contracts, calculating an interpolated forward point from the two corresponding contracts (Note: We can get the data for the forward point from Bloomberg directly or calculate the forward with the interest rates $r_{priceccy}$ and $r_{baseccy}$. We prefer the first method, as it probably is a more realistic measure because it includes the cross-currency basis.).
- S: Spot price (amount of price currency per unit of base currency); from EBS
- C, P: Premium of the call (C) and put (P) option; from DTCC data

Using the above variables and formulas, we see that the only unknown quantity in Eq. (B.1), for a call option, or in Eq. (B.2) for a put option, is the implied volatility σ (note that this is not the same sigma as in the main part of the paper; here it is the implied volatility), which is our objective. Thus, we only need to numerically minimise the following equation and estimate the implied volatility of each transaction.

$$\min_{\sigma} ||C_{Blackformula}(\sigma) - C_{observed\ \ in\ \ DTCC}|| \rightarrow \sigma_{reconstructed} \tag{B.8}$$

B.2. Determine the OMM's side behind each transaction

Once we estimate the implied volatility behind a transaction, we can match the transaction time stamp (which is given by the DTCC data) to a corresponding reference bid/ask price stream for the implied volatility (from Bloomberg). Note that it is important that the reference price stream is adjusted to match the tenor and delta of the reconstructed implied volatility of the transaction. Thus, the reference price stream onto which we match the reconstructed implied volatility of the transaction results from spline interpolation of the volatility surface (from Bloomberg) at the time when the transaction occurred. If the difference between the estimated implied volatility and the ask price of Bloomberg is *smaller* than the corresponding difference between the estimated implied volatility and the bid price from Bloomberg, we label the transaction as a paid, as the transaction probably occurred on the ask price. Conversely, if the difference between the estimated implied volatility and the ask price from Bloomberg is *larger* than the corresponding difference between the estimated implied volatility and the bid price from Bloomberg, we label the transaction as a given.

$$|\sigma_{estimated} - \sigma^{ask}_{reference}| - |\sigma_{estimated} - \sigma^{bid}_{reference}| \rightarrow \begin{cases} < 0, & \text{paid} \\ > 0, & \text{given} \end{cases} \tag{B.9}$$

Then, as the OMM is always liquidity provider, she is short the position if the transaction is a paid and long if the transaction is a given.

B.3. Quality of the reconstruction

Frömmel et al. (2020) present a nice discussion of the quality of trade reconstruction algorithms. They conclude that the modified EMO rule suggested by Chakrabarty et al. (2007) performs best, which works in the following manner: First, for trades at the quote and up to 30% below the ask or above the bid, the quote rule of Hasbrouck (1988) should be applied (quote rule: if price above (below) midpoint: buy (sell); Prices at midpoint unclassified; This method uses quote data.). Then, for all trades in the inner 40% of the spread, the tick rule of Blume et al. (1989) should be applied (tick rule: if price higher (lower) than previous: buy (sell); If no price change, same as previous. This method uses transaction data.). In our reconstruction, we only use the quote rule. The rationale for this decision is discussed in this subsection.

While the modified EMO rule is straightforward to apply for a centralized market, the application to the FX option market is a bit challenging, as the FX options market is a fragmented OTC market, so there is no one central exchange that records everything. There are rather several repositories to which the transactions that occur on exchanges or between two counterparties (OTC) are reported to. This means there is not one trade data set to use for the tick rule and not one reference price data set to use for the quote rule.

We take the Bloomberg best bid and ask price as reference quote price, which is an aggregated price from several market places. And as it is an aggregated best bid and ask price stream, the corresponding spread is small. This helps for the reconstruction, as most transactions have prices that lie outside of this tight spread (as most market participant cannot access those best bid and ask prices) and thus can be clearly identified as paid or given, according to the quote rule. With this reference price we apply the quote rule.

To discuss the quality of the reconstruction based on the quote rule, we calculate the difference of each trade price, $p_{i,t}$, from the contemporaneous far touch reference price, $p_t^{fartouch}$ and normalize it by the contemporaneous spread, $p_t^{ask} - p_t^{bid}$. If the trade is a paid, the far touch is the offer and if the trade is a given, the far touch is the bid. D_i is a $+1$ if the transaction is a given and -1 if the transaction is a paid. Thus m_i tells us how many spreads away the transaction occurred from the far touch.

$$m_i = D_i \frac{\left(p_{i,t} - p_t^{fartouch}\right)}{p_t^{ask} - p_t^{bid}} \tag{B.10}$$

This is what we see in Fig. B.7. A positive value indicates that the trade occurred in the spread, while a negative value indicates that the trade occurred outside of the spread. As expected, most trades occur outside of the tight Bloomberg spread.

The modified EMO rule assigns all trades at and up to 30% below the ask or above the bid to the corresponding far touch according to the quote rule (so it assigns a paid if the far touch is the ask and it assigns a given if the far touch is the bid). Only the transactions that lie in the inner 40% of the spread need to be reassigned using the tick rule. In our case, this is only amounts to 16% of all EURUSD transactions and 19% of all USDJPY transactions. Thus, for only a minor share of our transactions the modified EMO rule actually requires further analysis.

And as discussed above, using the tick rule is now a bit more tricky, as we do not have a complete set of transactions, but only the transactions reported to DTCC. And since applying the tick rule on an incomplete set of transactions is not ideal, we stick to the pure quote rule for our classification.

Nevertheless, for the sake of completeness we applied the tick rule (on the DTCC data set) to analyze how many trades would be classified differently when applying the tick rule (instead of the quote rule) to the trades that occurred in the inner 40% of the spread, as suggested by the modified EMO rule. We find that for EURUSD only 6% and for USDJPY only 7% of the trades in the inner 40% would actually be classified differently. In total, this means using the modified EMO rule vs. the quote rule impacts less than 1% of all transaction of EURUSD and 1.3% of all transaction of USDJPY. In other words, the quote rule and the modified EMO rule is identical for around 99% of all transactions in our data set.

Thus, as applying the tick rule on an incomplete set of transactions is not ideal and as it further does not even impact the outcome, we use the quote rule for the reconstruction.

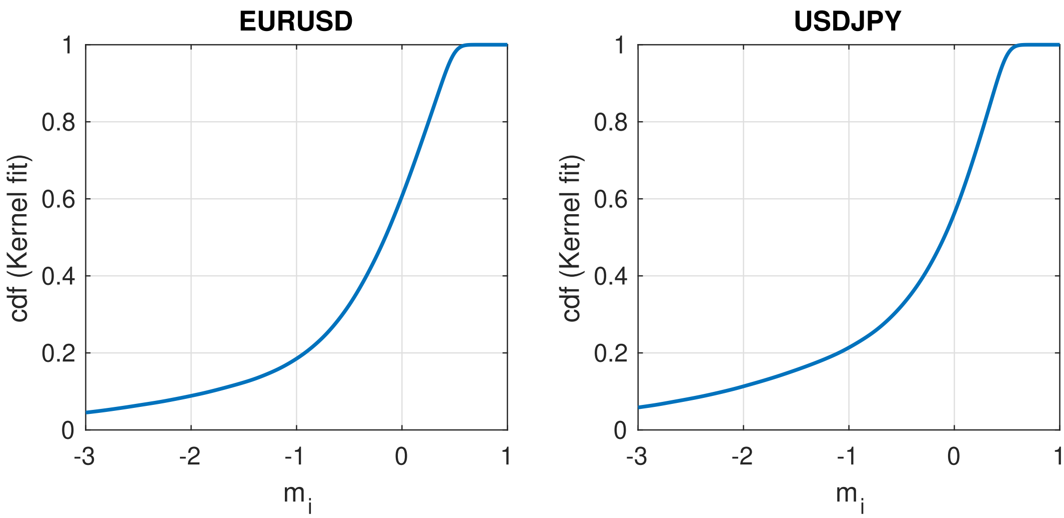

Fig. B.7. Cumulative distribution function (cdf) of the variable as defined Eq. B.10. m_i tells us how many spreads away the transaction occurred from the far touch. A positive value indicates that the trade occurred in the spread, while a negative value indicates that the trade occurred outsid.e of the spread.

Appendix C. Determine the OMM's delta and gamma exposure at each time step

Now that we have derived the OMM's position behind each transaction, we can calculate her potential delta and gamma exposure at each point in time using Eqs. (C.1) and (C.2) for the delta and Eq. (C.3) for the gamma. K is the strike price, S is the current spot price, F is the forward price for the remaining time, σ is the implied volatility for the remaining time, and τ is the remaining time.

$$\Delta_{call} = \frac{\partial C(F, K, S, \sigma, \tau)}{\partial S} = \Phi(d_+) \tag{C.1}$$

$$\Delta_{put} = \frac{\partial P(F, K, S, \sigma, \tau)}{\partial S} = -\Phi(-d_+) \tag{C.2}$$

$$\Gamma = \frac{\phi(d_1)}{S\sigma\sqrt{\tau}} \tag{C.3}$$

Note that ϕ is the standard normal probability density function and Φ is the cumulative distribution function of the standard normal distribution. Both measures, gamma and delta, are subsequently multiplied by the signed notional of each position (e.g., -1 million base currency if the OMM is short an option with a notional of 1 million and $+1$ million base currency if the OMM is long an option with a notional of 1 million), resulting in the gamma and delta exposure of the OMM. Eq. (C.4) describes the delta exposure and Eq. (C.5) the gamma exposure, whereby N is the notional of the option contract. If we sum up the gamma exposures of all active (not yet expired) positions for a certain point in time, we end up with the net gamma exposure of the OMM for this time.

$$\delta_{exposure} = \Delta * N \tag{C.4}$$

$$\gamma_{exposure} = \Gamma * N \tag{C.5}$$

$$sign(D) = \begin{cases} 1 & \text{if market maker is long} \\ -1 & \text{if market maker is short} \end{cases} \tag{C.6}$$

Appendix D. Is our model tradable?

Note that our non-linear regression used for the model estimation relates the OMM's gamma exposure in the beginning of the time period to the realised volatility during the time period. Thus, our regression suggests that based on the publicly available trade repository data we use in this report (and the subsequent estimation of the OMM's gamma exposure), a good forecast of the realised spot market volatility is possible. As the realised volatility is strongly correlated with the implied volatility, this could suggest a potential trading strategy. If we see that the OMM increased her short gamma exposure by i.e., $10bn$ EURUSD from time $t-2$ to time $t-1$ (so the net gamma exposure changes by $-10bn$), we know from our model parameter estimates that the realised spot market volatility of EURUSD in the time period $t-1$ to t will go up by $\frac{7.2\%}{1+(-10bn)\cdot\left(0.00009\frac{\%}{bn}\right)} - 7.2\% = 0.007\%$. Furthermore, we know that the realised spot market volatility is correlated with the implied volatility. If we therefore buy an option at $t-1$ and sell the option again at t, we sell 0.007% higher (option prices are usually quoted in terms of volatility). The converse is true if we see that the OMM decreases her short gamma exposure by 10*bn*. Of course, this trading strategy is based on an isolated view that only takes the observed feedback effect that the OMM gamma exposure has on the spot market volatility into account. We empirically tested this trading strategy with the data behind the results presented in Table 1 and found that the observed effect has a Sharpe ratio close to zero, which indicates efficient market behaviour. Additionally, the bid ask spreads in the option markets are usually wider than 0.007% anyway, so taking transaction costs into account, the strategy would even be unprofitable.

Appendix E. Estimation with intraday data

In this section we discuss the feedback effect on the spot market volatility independent of our model framework on an intraday basis.

E.1. Estimation of a random intercept linear mixed-effect model

As we can reconstruct the gamma exposure of the OMM on an intraday frequency, we can also quantify the effect with this intraday data. One difficulty of this intraday estimation is that the fundamental volatility (the constant of the linear regression model) depends on the intraday time, as it is influenced by the intraday liquidity cycle (trades influence prices and therefore also the volatility). We therefore cannot just mix the data from different intraday times and estimate the linear regression model. Thus, we use a mixed-effect model, namely, a random intercept linear mixed-effect model, as presented in Eq. E.1. As when we presented the gamma exposure, each trading day is split into 12 two-hour intervals. [11] We define a dummy variable for the corresponding intraday time stamp, I_m (where $m = \{1, \ldots, 12\}$), that indicates the corresponding intraday interval. I_m then is our grouping variable for the constant $\alpha_1[\%]$, as the fundamental volatility depends on the intraday time stamp. This construct, the constant conditional on the grouping variable, $(\alpha_1[\%]|I_m)$, is called the random effect, while the fixed effect is given by $\alpha_1[\%]$and $\alpha_2\left[\frac{\%}{\mathrm{bn}}\right]$. As the volatility can have extreme values on an intraday level, due to e.g., data releases, we remove outliers from the intraday data set. [12]

$$\sigma_{t,m}[\%] = \alpha_1[\%] + (\alpha_1[\%]|I_m) + \alpha_2\left[\frac{\%}{\mathrm{bn}}\right] \cdot N_t S_{t,m} \gamma^M_{t,m}[\mathrm{bn}] + \epsilon_{t,m} \tag{E.1}$$

Table E.2 displays the results of the estimation of Eq. E.1. Looking at the fixed effect, we see that the results confirm those obtained with the daily data. Note that now the fundamental volatility is nearly equal between EURUSD and USDJPY, as we account for the different intraday volatility regimes.

E.2. Relation to the volatility model

Note that the regression model presented in the previous section is just the linear approximation of Eq. 2.10. Eq. E.2 shows the derivation of the linear regression model, where we first write the series expansion and then a linear approximation of it (the last step in the derivation is justified by $|\beta h^{OMM}_{net} N_t S_t \gamma^M_t| <= 1$).

$$\sigma_t = \frac{\nu}{\left(1 + \beta h^{OMM}_{net} N_t S_t \gamma^M_t\right)} \tag{E.2a}$$

$$= \nu \sum_{k=0}^{\infty} \left(-\beta h^{OMM}_{net} N_t S_t \gamma^M_t\right)^k \tag{E.2b}$$

$$= \nu\left(1 - \beta h^{OMM}_{net} N_t S_t \gamma^M_t + O(2)\right) \tag{E.2c}$$

$$\approx \underbrace{\nu}_{=\alpha_1} + \underbrace{\left(-\nu\beta h^{OMM}_{net}\right)}_{=\alpha_2} N_t S_t \gamma^M_t \tag{E.2d}$$

Therefore, the estimations presented above provide support for our volatility model or, rather, its linear approximation.

$N_t S_t \gamma^M_t$ is the gamma exposure of the OMM. As $\alpha_2 = -\nu\beta h^{OMM}_{net}$, where h^{OMM}_{net} can be positive or negative (depending on whether the OMM or the OMT hedges a larger share of their exposure), $\nu > 0$ (fundamental volatility) and $\beta > 0$ (market impact), the sign of α_2 tells us whether the OMM hedges a larger share of her delta exposure compared to the OMT (as

[11] UTC 00:00–02:00, 02:00–04:00, 04:00–06:00, 06:00–08:00, 08:00–10:00, 10:00–12:00, 12:00–14:00, 14:00–16:00, 16:00–18:00, 18:00–20:00, 20:00–22:00, 22:00–24:00

[12] We remove the highest and lowest 2.5 percentiles.

Table E.2
This table shows the regression results of the random intercept linear mixed-effect model presented in Eq. E.1 for EURUSD and USDJPY. t-statistics are presented in brackets. The stars *** denote significance at the 1% level.

	EURUSD	USDJPY
α_1 (%)	5.7***	6.2***
	(10.8)	(11.1)
α_2 ($\frac{\%}{\text{bn}}$)	−0.0007***	−0.0008***
	(-8.6)	(-3.3)
R-squared	0.34	0.17
F-test vs. const. model	74.5***	10.8***

explained before). We estimate α_2 to be negative, so we can conclude that $h_{net}^{OMM} > 0$ and therefore that the OMM hedges a higher share of her delta exposure compared to the OMT. The value of α_2 quantifies the percentage increase in spot market volatility per gamma exposure of the OMM.

As we estimate $\alpha_1 = \nu$ separately, we could calculate βh_{net}^{OMM} from $\alpha_2 = -\nu\beta h_{net}^{OMM}$. Nevertheless, we cannot further dissect βh_{net}^{OMM} in its parts, as β and h_{net}^{OMM} are not further specified.

E.3. What can we learn from the intraday estimation

First, we want to highlight that the results presented above are very much in line withe the model validation results presented in Section 4. This is further strong evidence that the quantification of the feedback effect works well with our reconstructed OMM data set.

So the quantification of the feedback effect of the OMM's gamma exposure onto the spot market volatility is given by the parameter α_2 of the intraday estimation (Eq. E.1).

The significance of α_2 tells us that the gamma exposure of the OMM actually has relevance for the spot market volatility, and its value tells us what the corresponding linear effect is per billion gamma exposure of the OMM. Concretely, we see that the overall feedback effect stemming from the OMM's gamma exposure, which is given by the value of α_2, suggests that a short gamma exposure of the OMM of -1 billion in the base currency results in an increase of the realised spot market volatility of approximately 0.001% for both EURUSD and USDJPY. This does not look like a large effect, but as we see from the data displayed in Fig. 3, the gamma exposure of the OMM is of order -10^3 billion in both EURUSD and USDJPY. Consequently, the resulting effect is actually an increase in the spot market volatility of approximately 1% $(-0.001\,[\frac{\%}{\text{bn}}] \cdot -10^3[bn] = 1[\%])$. Note that the feedback effect of the gamma exposure on the spot market volatility is (most probably) not linear, as we know from our volatility model. Therefore, the quantification of the feedback effect presented in this section is only a linear approximation.

The constant α_1 is also found to be highly significant. α_1 can be interpreted as the spot market volatility in the absence of any delta hedging activity, the fundamental volatility. Note that for USDJPY, we find a strong difference in the estimate of ν from the model validation Section 4, in which we used daily data, compared to the intraday USDJPY estimate presented above. The spot market volatility in the absence of any delta hedging activity is found to be lower for the daily estimate (5.0%) compared to the intraday estimate (6.2%). This difference stems from the fact that the volatility data for the daily estimate for USDJPY from the model validation Section 4 originate from the Asian trading hours, in which the volatility is usually lower than during the European or American trading hours. As the intraday model takes the intraday time explicitly into account (and therefore also the intraday volatility regime), and corrects for it with a random effect, we obtain a more meaningful estimate of α_1 for USDJPY in the intraday estimation.

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